AIRCRAFT AND SYSTEM TESTING ESTABLISHMENT

FLIGHT TEST AND EVALUATION REPORT

Mi-17 V5 — General Flight Characteristics and Level-Flight Evaluation

Document field	Details
Report number	FTER/TRG/MI17V5/001 (Illustrative)
Aircraft type	Mi-17 V5 medium-lift helicopter
Evaluation category	General flight characteristics
Document status	Training template — not an approved flight-test report
Aircraft serial / sortie / date	To be completed from authorized records
Prepared / reviewed / approved by	To be completed by designated personnel
Classification	To be assigned under applicable policy

Important: This document is a training and documentation template. It does not contain measured sortie results, establish aircraft-specific operating limits, or constitute an operational procedure, airworthiness determination, or formal acceptance decision.

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1. Summary

The Mi-17 V5 is a twin-engine, medium-lift helicopter operated by the Indian Air Force. Public descriptions identify cockpit and avionics features and describe roles including transport, air maintenance, search and rescue, medical evacuation, and firefighting. The exact configuration of an individual aircraft must be verified against authorized technical records.

This report provides a framework for documenting general flight characteristics, with emphasis on steady level flight. No actual sortie records or measured flight-test data have been supplied; therefore, all results remain pending and no aircraft clearance or formal acceptance decision is made.

2. Aim

To establish a documentation framework for evaluating general flight characteristics during a representative steady-flight condition, using validated records and applicable authorized criteria.

3. Scope

This report covers aircraft identification and configuration, environmental and instrumentation records, general principles of helicopter flight, data-quality assessment, uncertainty, findings, and recommendations. It does not prescribe aircraft-specific manoeuvres, operating limits, crew procedures, or acceptance criteria.

4. Aircraft Description

Feature	General description
Aircraft category	Medium-lift helicopter
Configuration	Twin-engine, main-rotor and tail-rotor helicopter
Propulsion	Twin turboshaft engines
Cockpit and avionics	Glass-cockpit instrumentation and integrated navigation systems, subject to aircraft configuration
Weather equipment	Public descriptions report onboard weather radar; verify aircraft fit
Night operations	Public descriptions report compatibility with night-vision equipment; verify aircraft fit
General roles	Transport, air maintenance, rescue, evacuation, and firefighting

The precise configuration, modifications, installed equipment, and serviceability of an individual helicopter must be confirmed from authorized records. Public descriptions do not establish the condition of a particular airframe.

5. Evaluation Objectives

• Establish aircraft identity and configuration traceability.

• Verify completeness and consistency of recorded flight parameters.

• Determine whether observations support a steady-flight classification.

• Assess consistency among available flight-instrument and engine-monitoring records.

• Document data-quality issues, uncertainty, and limitations.

• Determine whether the evidence is sufficient for a defensible technical conclusion.

6. Conditions Relevant to the Evaluation

6.1 Aircraft configuration

Parameter	Required record
Aircraft serial number	Authorized aircraft record
Modification standard	Verified configuration
Engine configuration	Applicable technical records
Aircraft mass and loading	Recorded authorized values
Centre of gravity	Verified loading documentation
External equipment	Configuration record
Maintenance status	Relevant serviceability records
Instrumentation configuration	Installed and calibrated equipment

6.2 Environmental conditions

Parameter	Information required
Date and time	Timestamp and time reference
Airfield / test area	Authorized record
Temperature and pressure	Recorded values and units
Wind conditions	Available meteorological observations
Visibility and cloud	Recorded observations
Turbulence	Qualitative assessment or measurement
Data source	Instrument or meteorological record

6.3 Instrumentation and data quality

Document the provenance, units, calibration status, resolution, and uncertainty of each measurement. Relevant data categories may include flight-instrument records, aircraft attitude, navigation data, engine-monitoring indications, environmental observations, and time synchronization. The actual instrumentation fit and processing requirements must be established from authorized documentation.

7. Methodology — General Steady-Flight Assessment

7.1 Conceptual basis

For a helicopter in steady, approximately level flight, the net vertical force is approximately zero. A simplified relationship is:

$$F_{\text{vertical, net}} \approx 0$$

This is a general physical relationship, not a measured result for the Mi-17 V5. Actual helicopter flight involves interacting aerodynamic forces, rotor dynamics, propulsion, aircraft attitude, and control-system behaviour.

7.2 Evaluation stages

Stage 1 — Record verification

Confirm aircraft identity, configuration, maintenance documentation, and instrumentation status.

Stage 2 — Data integrity

Check source records for missing observations, inconsistent units, invalid timestamps, and recording anomalies.

Stage 3 — Technical assessment

Examine validated records for consistency, variability, uncertainty, and agreement with applicable criteria.

Stage 4 — Reporting

Document observations, interpretations, unresolved issues, and evidence supporting each conclusion.

8. Analysis and Results

Assessment status: Pending validated flight-test data. The tables below are blank reporting templates, not actual aircraft results.

8.1 General assessment matrix

Assessment area	Evidence required	Result
Flight-state consistency	Validated time histories	Pending
Altitude consistency	Recorded instrument data	Pending
Attitude consistency	Relevant flight-instrument records	Pending
Engine indication consistency	Recorded engine-monitoring data	Pending
Navigation data integrity	Source and synchronization checks	Pending
Instrument reliability	Calibration and maintenance records	Pending
Data repeatability	Comparable validated observations	Pending

8.2 Results recording sheet

Parameter	Unit	Recorded result
Pressure altitude	ft or m	Not supplied
Airspeed	As defined by source instrument	Not supplied
Aircraft attitude	Degrees	Not supplied
Outside air temperature	°C	Not supplied
Aircraft mass	kg	Not supplied
Engine indications	Source-defined units	Not supplied
Observation duration	s or min	Not supplied
Measurement uncertainty	Parameter-specific	Not supplied

8.3 Interpretation of observations

Observation: what the recorded data directly demonstrate. **Interpretation:** what the evidence suggests within stated assumptions. **Conclusion:** what can be supported after data quality, uncertainty, and applicable criteria are considered. A stable altitude record alone is insufficient to establish the overall flight state or the serviceability of an aircraft system.

8.4 Proposed figures

• Altitude versus time.

• Recorded airspeed versus time.

• Aircraft attitude versus time.

• Engine-monitoring parameters versus time.

• Data-quality and measurement-uncertainty plots.

Each figure should identify its source, units, observation period, and processing notes. Simulated data used for classroom demonstrations must be clearly labelled as simulated.

9. Limitations and Error Considerations

Limitation	Potential effect
Instrument calibration uncertainty	Limits confidence in measurements
Inconsistent timestamps	Can distort comparisons between data sources
Missing observations	Can prevent a complete assessment
Environmental variability	Can affect comparisons between records
Configuration differences	Can invalidate direct comparisons
Data-processing errors	Can introduce misleading trends
Incomplete acceptance criteria	Prevents a formal pass/fail determination

The final report should include an uncertainty assessment based on actual instrumentation specifications and approved engineering methods.

10. Findings

Publicly available information describes the Mi-17 V5 as a medium-lift helicopter equipped with cockpit and navigation systems and designed for a range of transport and support roles. The information available for this report does not establish the measured performance or technical condition of any individual helicopter. No flight-test result, compliance determination, or system acceptance finding can therefore be declared.

Outstanding items include aircraft configuration and serviceability records, validated flight-test observations, instrument calibration and uncertainty documentation, applicable authorized acceptance criteria, and technical review.

11. Conclusion

This report provides a structured framework for documenting the general flight characteristics of the Mi-17 V5, including data-quality and reporting requirements relevant to a steady-flight evaluation. A formal technical conclusion must remain pending until the required evidence has been reviewed. This document is not an operational flight manual, approved flight-test card, or aircraft release-to-service document.

12. Recommendations

• Maintain traceability between reported results and source records.

• Document aircraft configuration and environmental conditions.

• Record uncertainty and data-quality limitations.

• Clearly identify illustrative material and unverified results.

• Obtain review and approval from the designated technical authority before issuing an official report.

13. Appendices

• Appendix A — Aircraft identification and configuration record.

• Appendix B — Source-data register.

• Appendix C — Instrumentation and calibration register.

• Appendix D — Data-quality and uncertainty assessment.

• Appendix E — Figures and supporting observations.

• Appendix F — Review comments and approval record.

References

1. Ministry of Defence, Government of India — Mi-17 V5 helicopter induction announcement: PIB release.
2. Ministry of Defence, Government of India — Multi-Purpose Mi-17 V5 Helicopter Inducted into Air Force: PIB release.
3. Department of Defence Production — Annual Report 2017–18: PDF report.